

Project Plan

A Modular Deep Learning Framework for Lightweight Image Recognition
CS5242 Group 09 Project

1. Project Motivation

Image recognition is a core computer vision task. Instead of only reproducing a standard model, our project studies how several simple plug-and-play modules can improve a lightweight baseline in a controlled and interpretable way. This makes the project suitable for a team setting: the baseline can be kept simple and reproducible, while each team member studies one concrete architectural improvement.

2. Project Description

We will build a lightweight image recognition system on the **Oxford-IIIT Pet** dataset. We will first implement a simple baseline, with complete data preprocessing, training, validation, testing, and evaluation. We will then add three independent modules: Module 1, Module 2, and Module 3. Each module will be tested individually and compared fairly under the same setting.

3. Proposed Solution

The project contains one baseline and three modular extensions:

- **Baseline:** implement the simplest working model and full training pipeline.
- **Module 1 :** study its motivation, add it to the network, and evaluate its effect.
- **Module 2 :** implement a second improvement and compare it with the baseline.
- **Module 3 :** implement a third module and analyze its benefit and cost.
- **Integration:** compare the baseline, each single-module version, and the best combined model.

Evaluation will include standard quantitative metrics, learning curves, and concise ablation studies. The final goal is a clear comparison showing how each module affects performance.

4. Expected Deliverables

- a reproducible baseline model and training pipeline;
- three independently implemented improvement modules;
- ablation results for each module and one final comparison table;
- runnable notebook with results
- presentation slides.

5. Project Milestones

Week	Milestone
Week 1	Finalize dataset, metrics, repository, and environment.
Week 2	Finish the baseline and the complete train/test pipeline.
Week 3	Complete Module 1 and Module 2 with preliminary experiments.
Week 4	Complete Module 3 and all ablation studies.
Week 5	Integrate the best model and organize figures/tables.
Week 6	Finalize the report, slides, and rehearsal.

References, if any, can be attached as extra pages.

Team Contract / Agreement

A Modular Deep Learning Framework for Lightweight Image Recognition

6. Team Members

Member	Name
Member 1	Wang Jiayu(A0326769L)
Member 2	Tong Yicheng(A0326416H)
Member 3	Lin Zhenghong(A0328419X)
Member 4	Deng Zhiyang(A0328988B)

7. Task Allocation

Member	Assigned Responsibilities
Member 1	Build the baseline model and the full experimental pipeline, including data loading, preprocessing, training, validation, testing, logging, model integration, and slides/presentation coordination.
Member 2	Take full ownership of Module 1, including motivation, implementation, debugging, ablation, experimental analysis, and report writing for this module.
Member 3	Take full ownership of Module 2, including motivation, implementation, debugging, ablation, experimental analysis, and report writing for this module.
Member 4	Take full ownership of Module 3, including motivation, implementation, debugging, ablation, experimental analysis, and report writing for this module.

8. Shared Responsibilities

To ensure balanced contribution, all members will also:

- join planning meetings and weekly progress checks;
- review teammates' code and help with debugging/integration;
- discuss experiment settings, results, and final conclusions together;
- contribute to report revision and presentation rehearsal.

9. Team Agreement

We agree that:

1. each member will contribute fairly and complete assigned tasks on time;
2. risks or delays will be communicated early to the team;
3. major design choices and final conclusions will be discussed jointly;
4. the final report and codebase may be refined collaboratively for consistency.

10. Signatures

Name	Signature	Date
WANG JIAYU		08 March 2026
TONG YICHENG	Tong YC	08 March 2026
LIN ZHENGHONG	LIN ZHENGHONG	08 March 2026
DENG ZHIYANG	Deng Zhiyang	08 March 2026